```
>> dynare mcmc.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Substitution of exo leads: added 4 auxiliary variables and equations.
Substitution of exo lags: added 1 auxiliary variables and equations.
Found 28 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 18 equation(s)
                                 and 18 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
7 block(s) found:
  5 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 18 equation(s)
                                 and 18 feedback variable(s).
Preprocessing completed.
Preprocessing time: 0h00m00s.

STEADY-STATE RESULTS:

q                          32.8507
k                          4.41224
R                          1.01351
b                          143.013
x                          0.0406367
varphi                     0.0303425
mu                         0.00686924
kp                         2.20776
chi                        1.02177
nu                         0.00432944
eta                        1.28813
zeta                       1.02177
N                          41.8193
phi                        3.41978
Ne                         41.5333
Nn                         0.286025
xp                         6.22046
bp                         -101.193
Y                          3.45318
eb                         0
N_obs                      0
Y_obs                      0
q_obs                      0

EIGENVALUES:
         Modulus             Real        Imaginary
               0                0                0
       1.385e-22       -1.385e-22                0
        2.66e-17         2.66e-17                0
```

```
        3.6e-16        -3.6e-16               0
      4.774e-16       4.774e-16               0
      4.146e-08      -2.152e-08       3.544e-08
      4.146e-08      -2.152e-08      -3.544e-08
      4.303e-08       4.303e-08               0
         0.1469          0.1469               0
         0.9804          0.9804               0
          1.018           1.018               0
           1.02            1.02               0
          1.022           1.022               0
          1.038           1.038               0
          3.365          -3.365               0
      8.783e+18      -8.783e+18               0
      5.524e+35      -5.524e+35               0
      5.897e+35      -5.897e+35               0
      1.921e+38       1.921e+38               0
            Inf             Inf               0

There are 10 eigenvalue(s) larger than 1 in modulus for 10 forward-looking variable(s)
The rank condition is verified.



======== Identification Analysis ========

Testing prior mean
-----------
The model does not solve for prior_mean (info = 4: Blanchard & Kahn conditions are not satisfied:
indeterminacy.)
-----------
Try sampling up to 50 parameter sets from the prior.
Found a random draw from the priors that solves the model:
    0.0066    0.0160    0.0078    3.5377    5.5017

Identification now continues for this draw.
Note that differences in the criteria could be due to numerical settings,
numerical errors or the method used to find problematic parameter sets.
Settings:
    Derivation mode for Jacobians:                         Analytic using sylvester equations
    Method to find problematic parameters:                 Nullspace and multicorrelation
coefficients
    Normalize Jacobians:                                   Yes
    Tolerance level for rank computations:                 robust
    Tolerance level for selecting nonzero columns:         1e-08
    Tolerance level for selecting nonzero singular values: 1e-03

REDUCED-FORM:
    All parameters are identified in the Jacobian of steady state and reduced-form solution
matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):
    All parameters are identified in the Jacobian of steady state and minimal system
(rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):
    All parameters are identified in the Jacobian of mean and spectrum (rank(Gbar) is full with
tol = robust).
```

```
MOMENTS (ISKREV, 2010):
    All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol
= robust).

==== Identification analysis completed ====


You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): -708.0972
-----------------
f at the beginning of new iteration,       708.0972278947
Predicted improvement:  1897518.430577544
lambda =          1; f =          996.5163990
lambda =    0.33333; f =          724.6145403
lambda =    0.11111; f =           59.7721335
lambda =   0.037037; f =          -83.3504661
lambda =   0.012346; f =         -198.7676946
lambda =  0.0041152; f =         -260.7238603
lambda =  0.0013717; f =         -227.8997269
lambda = 0.00045725; f =          -16.8127878
Norm of dx     19.481
----
Improvement on iteration 1 =      968.821088170
-----------------
f at the beginning of new iteration,      -260.7238602753
Predicted improvement:      219.855642380
lambda =          1; f =         -260.7074740
lambda =    0.33333; f =         -260.7227767
lambda =    0.11111; f =         -260.7238588
lambda =   0.037037; f =         -269.3812987
Norm of dx     0.2097
----
Improvement on iteration 2 =        8.657438402
-----------------
f at the beginning of new iteration,      -269.3812986768
Predicted improvement:      213.685350211
lambda =          1; f =         -269.3025554
lambda =    0.33333; f =         -269.3736760
lambda =    0.11111; f =         -269.3807755
lambda =   0.037037; f =         -269.3812967
lambda =   0.012346; f =         -273.2687735
lambda =   0.023866; f =         -262.1719272
lambda =    0.01607; f =         -272.9219770
Norm of dx    0.29134
----
Improvement on iteration 3 =        3.887474844
-----------------
f at the beginning of new iteration,      -273.2687735205
Predicted improvement:        0.800973945
lambda =          1; f =         -274.2143827
Norm of dx  0.0019826
----
Improvement on iteration 4 =        0.945609146
-----------------
f at the beginning of new iteration,      -274.2143826666
Predicted improvement:        1.005968658
lambda =          1; f =         -273.5115429
lambda =    0.33333; f =         -274.6158058
```

```
Norm of dx  0.0035248
----
Improvement on iteration 5 =        0.401423108
-----------------
f at the beginning of new iteration,      -274.6158057744
Predicted improvement:        0.030130573
lambda =          1; f =         -274.6610968
lambda =     1.9332; f =         -274.6769988
Norm of dx 0.00038329
----
Improvement on iteration 6 =        0.061193068
-----------------
f at the beginning of new iteration,      -274.6769988427
Predicted improvement:        0.029815726
lambda =          1; f =         -274.7352353
lambda =     1.9332; f =         -274.7870440
lambda =     3.7372; f =         -274.8804207
lambda =     7.2247; f =         -275.0365671
lambda =     13.967; f =         -275.2536739
Norm of dx  0.0005967
----
Improvement on iteration 7 =        0.576675034
-----------------
f at the beginning of new iteration,      -275.2536738764
Predicted improvement:        0.751759965
lambda =          1; f =         -276.5106552
lambda =     1.9332; f =         -277.1120257
Norm of dx   0.018355
----
Improvement on iteration 8 =        1.858351865
-----------------
f at the beginning of new iteration,      -277.1120257416
Predicted improvement:        0.518852974
lambda =          1; f =         -277.8585301
lambda =     1.9332; f =         -277.8997825
Norm of dx    0.01286
----
Improvement on iteration 9 =        0.787756713
-----------------
f at the beginning of new iteration,      -277.8997824550
Predicted improvement:        0.602968643
lambda =          1; f =         -278.2281984
lambda =    0.33333; f =         -278.2061083
lambda =    0.64439; f =         -278.3172226
Norm of dx   0.013689
----
Improvement on iteration 10 =        0.417440191
-----------------
f at the beginning of new iteration,      -278.3172226461
Predicted improvement:        0.022350910
lambda =          1; f =         -278.3362404
Norm of dx  0.0022367
----
Improvement on iteration 11 =        0.019017741
-----------------
f at the beginning of new iteration,      -278.3362403872
Predicted improvement:        0.002972176
lambda =          1; f =         -278.3404299
```

```
lambda =     1.9332; f =         -278.3411514
Norm of dx 0.00064613
----
Improvement on iteration 12 =        0.004911048
-----------------
f at the beginning of new iteration,      -278.3411514356
Predicted improvement:        0.003081775
lambda =          1; f =         -278.3471349
lambda =     1.9332; f =         -278.3523870
lambda =     3.7372; f =         -278.3616376
lambda =     7.2247; f =         -278.3761747
lambda =     13.967; f =         -278.3919643
Norm of dx  0.0010236
----
Improvement on iteration 13 =        0.050812823
-----------------
f at the beginning of new iteration,      -278.3919642583
Predicted improvement:        0.052267707
lambda =          1; f =         -278.4947479
lambda =     1.9332; f =         -278.5875382
lambda =     3.7372; f =         -278.7586682
lambda =     7.2247; f =         -279.0604098
lambda =     13.967; f =         -279.5459073
lambda =         27; f =         -280.1755930
Norm of dx   0.022908
----
Improvement on iteration 14 =        1.783628785
-----------------
f at the beginning of new iteration,      -280.1755930434
Correct for low angle: 0.00318057
Predicted improvement:        3.015604108
lambda =          1; f =         -285.0288727
lambda =     1.9332; f =         -278.9429235
lambda =     1.3017; f =         -284.6979416
Norm of dx    0.97125
----
Improvement on iteration 15 =        4.853279692
-----------------
f at the beginning of new iteration,      -285.0288727354
Predicted improvement:        0.892243950
lambda =          1; f =         -284.4740855
lambda =    0.33333; f =         -285.4313139
Norm of dx    0.32003
----
Improvement on iteration 16 =        0.402441126
-----------------
f at the beginning of new iteration,      -285.4313138611
Predicted improvement:        0.560297633
lambda =          1; f =         -284.3069499
lambda =    0.33333; f =         -285.6029225
Norm of dx    0.24803
----
Improvement on iteration 17 =        0.171608608
-----------------
f at the beginning of new iteration,      -285.6029224689
Predicted improvement:        0.013687411
lambda =          1; f =         -285.6149314
Norm of dx   0.036974
```

```
----
Improvement on iteration 18 =        0.012008912
-----------------
f at the beginning of new iteration,      -285.6149313813
Predicted improvement:        0.000610361
lambda =          1; f =         -285.6154792
Norm of dx  0.0049652
----
Improvement on iteration 19 =        0.000547776
-----------------
f at the beginning of new iteration,      -285.6154791569
Predicted improvement:        0.000052365
lambda =          1; f =         -285.6155584
lambda =     1.9332; f =         -285.6155878
Norm of dx 0.00050652
----
Improvement on iteration 20 =        0.000108612
-----------------
f at the beginning of new iteration,      -285.6155877688
Predicted improvement:        0.000077773
lambda =          1; f =         -285.6157378
lambda =     1.9332; f =         -285.6158703
lambda =     3.7372; f =         -285.6161062
lambda =     7.2247; f =         -285.6164859
lambda =     13.967; f =         -285.6169335
Norm of dx 0.00025597
----
Improvement on iteration 21 =        0.001345696
-----------------
f at the beginning of new iteration,      -285.6169334647
Predicted improvement:        0.001719941
lambda =          1; f =         -285.6202159
lambda =     1.9332; f =         -285.6230036
lambda =     3.7372; f =         -285.6276388
lambda =     7.2247; f =         -285.6337797
Norm of dx  0.0056804
----
Improvement on iteration 22 =        0.016846278
-----------------
f at the beginning of new iteration,      -285.6337797425
Correct for low angle: 0.00490646
Predicted improvement:        0.018907417
lambda =          1; f =         -285.6669581
lambda =     1.9332; f =         -285.6897587
lambda =     3.7372; f =         -285.7125286
Norm of dx    0.06432
----
Improvement on iteration 23 =        0.078748824
-----------------
f at the beginning of new iteration,      -285.7125285665
Correct for low angle: 0.00363207
Predicted improvement:        0.032136743
lambda =          1; f =         -285.7477719
Norm of dx   0.067761
----
Improvement on iteration 24 =        0.035243348
-----------------
f at the beginning of new iteration,      -285.7477719142
```

```
Predicted improvement:        0.001968639
lambda =          1; f =         -285.7495844
Norm of dx   0.019605
----
Improvement on iteration 25 =        0.001812499
-----------------
f at the beginning of new iteration,      -285.7495844129
Predicted improvement:        0.000018415
lambda =          1; f =         -285.7496040
Norm of dx 0.00058636
----
Improvement on iteration 26 =        0.000019566
-----------------
f at the beginning of new iteration,      -285.7496039788
Correct for low angle: 0.00423132
Predicted improvement:        0.000000018
lambda =          1; f =         -285.7496040
lambda =    0.33333; f =         -285.7496040
lambda =    0.11111; f =         -285.7496040
Norm of dx 7.8453e-05
----
Improvement on iteration 27 =        0.000000002
improvement < crit termination

Final value of minus the log posterior (or likelihood):-285.749604

MODE CHECK

Fval obtained by the optimization routine: -285.749604

RESULTS FROM POSTERIOR ESTIMATION
parameters
         prior mean     mode    s.d.  prior pstdev

gamma_q      3.0000   1.6131  0.6868   norm 1.0000
gamma_bY     0.0000   1.7946  0.3581   norm 3.0000

standard deviation of shocks
         prior mean     mode    s.d.  prior pstdev

eN           0.0100   0.0517  0.0053   invg 0.0050
eY           0.0100   0.0167  0.0016   invg 0.0050
eq           0.0100   0.0081  0.0011   invg 0.0050


Log data density [Laplace approximation] is 269.932516.

Estimation::mcmc: Multiple chains mode.
Estimation::mcmc: Old metropolis.log file successfully erased!
Estimation::mcmc: Creation of a new metropolis.log file.
Estimation::mcmc: Searching for initial values...
Estimation::mcmc: Initial values found!

Estimation::mcmc: Write details about the MCMC... Ok!
Estimation::mcmc: Details about the MCMC are available in mcmc/metropolis\mcmc_mh_history_0.mat


Estimation::mcmc: Number of mh files: 1 per block.
```

```
Estimation::mcmc: Total number of generated files: 2.
Estimation::mcmc: Total number of iterations: 125000.
Estimation::mcmc: Current acceptance ratio per chain:
                                                       Chain  1: 47.9592%
                                                       Chain  2: 47.76%
Estimation::mcmc: Total number of MH draws per chain: 125000.
Estimation::mcmc: Total number of generated MH files: 1.
Estimation::mcmc: I'll use mh-files 1 to 1.
Estimation::mcmc: In MH-file number 1 I'll start at line 25001.
Estimation::mcmc: Finally I keep 100000 draws per chain.



MCMC Inefficiency factors per block
Parameter       Block 1     Block 2
SE_eN            32.314      37.563
SE_eY            23.931      30.264
SE_eq            30.001      30.542
gamma_q          23.154      25.386
gamma_bY         28.229      37.258



Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for
chain 1.
p-values are for Chi2-test for equality of means.
Parameter         Post. Mean        Post. Std   p-val No Taper   p-val 4% Taper   p-val 8% Taper
p-val 15% Taper
SE_eN                  0.054            0.006            0.000            0.132
0.114           0.099
SE_eY                  0.017            0.002            0.516            0.870
0.862           0.849
SE_eq                  0.008            0.001            0.000            0.413
0.416           0.427
gamma_q                1.575            0.695            0.004            0.569
0.569           0.578
gamma_bY               1.948            0.398            0.001            0.523
0.510           0.517




Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for
chain 2.
p-values are for Chi2-test for equality of means.
Parameter         Post. Mean        Post. Std   p-val No Taper   p-val 4% Taper   p-val 8% Taper
p-val 15% Taper
SE_eN                  0.054            0.006            0.000            0.491
0.456           0.443
SE_eY                  0.017            0.002            0.078            0.752
0.748           0.741
SE_eq                  0.008            0.001            0.030            0.673
0.662           0.634
gamma_q                1.566            0.695            0.007            0.590
```

```
0.610            0.592
gamma_bY               1.951            0.390            0.000            0.150
0.188            0.244


Univariate convergence diagnostic, Brooks and Gelman (1998):
    Parameter 1...  Done!
    Parameter 2...  Done!
    Parameter 3...  Done!
    Parameter 4...  Done!
    Parameter 5...  Done!
marginal density: I'm computing the posterior mean and covariance...  Done!
marginal density: I'm computing the posterior log marginal density (modified harmonic mean)...
Done!


ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 270.089213.
parameters
           prior mean   post. mean        90% HPD interval    prior       pstdev
gamma_q         3.000       1.5721      0.4219      2.6970     norm       1.0000
gamma_bY        0.000       1.9539      1.3096      2.5735     norm       3.0000

standard deviation of shocks
           prior mean   post. mean        90% HPD interval    prior       pstdev

eN              0.010       0.0545      0.0448      0.0640     invg       0.0050
eY              0.010       0.0172      0.0145      0.0199     invg       0.0050
eq              0.010       0.0084      0.0065      0.0103     invg       0.0050
Total computing time : 0h10m43s
>>
```